

Brazilian E-Commerce Analytics Report

GitHub Repository

https://github.com/Phoenix881/Bubstal_Technical_Assessment

Executive Summary

This project analyzes the Olist Brazilian e-commerce marketplace dataset, covering approximately 100K orders from 2016 to 2018. The analysis focuses on business performance, monthly demand forecasting for the June-August 2018 holdout period, and LLM-assisted recommendations through OpenRouter.

Key results:

Metric	Value
Total orders	99,441
Total revenue	BRL 16,008,872.12
Average order value	BRL 160.99
Forecast model	seasonal_naive_trend_ensemble
Forecast RMSE	484.72 orders
Forecast MAE	447.00 orders
LLM model used	baidu/cobuddy-20260430:free

The marketplace shows strong revenue concentration in a small set of categories, heavy dependence on credit cards and boleto payments, and meaningful logistics improvement opportunities. The demand forecast slightly underpredicted the June-August 2018 holdout period, but produced a reasonable baseline for operational planning.

Data and Scope

The dataset contains Olist marketplace orders, order items, payments, reviews, customers, sellers, products, product category translations, and geolocation data.

Analysis window:

- Full observed order range: September 2016 to October 2018
- Forecast training cutoff: May 31, 2018
- Forecast holdout period: June 1, 2018 to August 31, 2018
- LLM context uses complete monthly revenue through August 2018 to avoid overinterpreting the very small September and October partial-month data.

Revenue is computed from `payment_value`, aggregated to order level before monthly and total calculations. Demand is measured as monthly unique order count.

Business Performance

Total marketplace revenue was BRL 16,008,872.12 across 99,441 orders, with an average order value of BRL 160.99.

Recent complete monthly revenue in the forecast holdout period:

Month	Revenue
2018-06	BRL 1,023,880.50
2018-07	BRL 1,066,540.75
2018-08	BRL 1,022,425.32

The top revenue categories are:

Category	Revenue	Orders
health_beauty	BRL 1,441,248.07	8,836
watches_gifts	BRL 1,305,541.61	5,624
bed_bath_table	BRL 1,241,681.72	9,417
sports_leisure	BRL 1,156,656.48	7,720
computers_accessories	BRL 1,059,272.40	6,689

These categories are the strongest candidates for inventory protection, campaign planning, and seller enablement because they combine high revenue with meaningful order volume.

Payment Mix

Payment value is concentrated in credit card and boleto:

Payment Type	Payment Value	Orders
credit_card	BRL 12,542,084.19	76,505
boleto	BRL 2,869,361.27	19,784
voucher	BRL 379,436.87	3,866
debit_card	BRL 217,989.79	1,528
not_defined	BRL 0.00	3

Credit card is the dominant method by value and order count. Boleto is also material, so checkout reliability and payment-status communication matter for Brazilian customer behavior.

Customer Experience and Logistics

Customer satisfaction is generally positive, but there is a visible low-score segment:

Metric	Value
Reviews	98,410

Metric	Value
Average review score	4.09
Share of scores 1-2	14.7%

Delivery performance:

Metric	Value
Delivered orders	96,476
Average delivery time	12.56 days
Late delivery share	8.1%

Logistics is one of the clearest improvement areas. Even with an average score of 4.09, the 14.7% share of low reviews suggests that reducing late deliveries and improving delivery communication could protect customer satisfaction.

Geographic Demand

The largest customer states by orders are:

State	Orders
SP	41,746
RJ	12,852
MG	11,635
RS	5,466
PR	5,045

Operational and marketing improvements should start in SP, RJ, and MG because those states contain the largest order concentrations.

Demand Forecast

The forecasting task predicts monthly order volume for June, July, and August 2018 using training data through May 31, 2018.

The model implemented is `seasonal_naive_trend_ensemble`. It blends:

- seasonal signal from the prior year when available
- recent trend from the last 12 months
- trailing moving average from recent demand

This model is intentionally simple and explainable, which fits the short assessment window and gives leadership a transparent baseline.

Forecast results:

Month	Predicted Orders	Actual Orders	Error
2018-06	5,456	6,167	-711
2018-07	5,956	6,292	-336
2018-08	6,218	6,512	-294

Evaluation:

Metric	Value
RMSE	484.72
MAE	447.00

The model underpredicted all three holdout months, especially June 2018. It is still directionally useful for capacity planning, but a production version should compare multiple model classes and include richer explanatory variables such as category mix, state-level demand, seller availability, holiday effects, and marketing seasonality.

LLM-Assisted Recommendations

The recommendation step uses OpenRouter's OpenAI-compatible chat completion API. The project loads `OPENROUTER_API_KEY` from `.env`, builds a structured analysis context, and requests JSON recommendations.

Model used successfully:

```
baidu/cobuddy-20260430:free
```

The final generated recommendations were:

Priority	Action	Rationale
high	Increase inventory and marketing spend for top revenue categories: health_beauty, watches_gifts, bed_bath_table, sports_leisure, computers_accessories	These categories generated BRL 1,441,248.07, BRL 1,305,541.61, BRL 1,241,681.72, BRL 1,156,656.48, and BRL 1,059,272.40 respectively, representing a significant portion of total revenue BRL 16,008,872.12.
high	Optimize logistics to reduce late delivery share of 8.1% and average delivery time of 12.56 days	Late deliveries may impact customer satisfaction, as 14.7% of reviews are scores 1-2; improving delivery performance could increase satisfaction.
medium	Enhance boleto checkout process to reduce friction	Boleto accounts for BRL 2,869,361.27 across 19,784 orders, making it the second-

Priority	Action	Rationale
		largest payment method by value and orders.

The code includes a deterministic fallback recommendation path. If OpenRouter is rate-limited, blocked by a provider, or returns malformed JSON, the project still returns usable recommendations and marks `model_used` with `_local_fallback`.

Limitations

- The September and October 2018 order counts are tiny, likely representing partial or anomalous periods, so the final LLM context focuses on complete revenue through August 2018.
- The forecast is a baseline model, not a fully tuned forecasting system.
- Revenue is based on payment values, while category revenue is based on item price plus freight value. These are useful but not identical business lenses.
- OpenRouter free models can be intermittent due to provider rate limits, region restrictions, and upstream availability.
- Review comments are in Portuguese and were not semantically analyzed in this version.

Reproducibility

Install dependencies:

```
python3 -m venv .venv
.venv/bin/python -m pip install -r requirements.txt
```

Download the dataset:

```
curl -L --fail -o data/olist_brazilian_ecommerce.zip
https://www.kaggle.com/api/v1/datasets/download/olistbr/brazilian-ecommerce
unzip -o data/olist_brazilian_ecommerce.zip -d data
```

Run tests:

```
.venv/bin/python -m pytest -v
```

Execute the notebook:

```
.venv/bin/jupyter nbconvert --execute --to notebook --inplace src/notebook.ipynb
--ExecutePreprocessor.timeout=240
```

Primary artifacts:

- `src/analysis.py`: analysis, forecasting, and recommendation functions
- `src/notebook.ipynb`: executed EDA notebook with visualizations
- `outputs/analysis_summary.json`: final structured summary
- `outputs/recommendations.json`: LLM recommendation output

- `outputs/monthly_demand.csv`: monthly order demand
- `outputs/revenue_by_month.csv`: monthly revenue